

FlowRecon-Bench: A benchmark for sparse-sensor flow-field reconstruction across flow regimes, with calibrated uncertainty

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Reconstructing a full flow field from sparse point sensors is a core task of experimental and operational fluid mechanics, and a rapidly growing machine-learning literature now addresses it with deterministic operator learners and conditional generative models. Yet the field has no shared benchmark: published methods are evaluated on incompatible, often temporally leaky splits, ensembles are drawn but never scored with proper probabilistic metrics, and classical baselines are rarely reported. We present FlowRecon-Bench, a benchmark spanning five flow regimes — laminar cylinder vortex shedding (2D, multi-Reynolds), two-dimensional Kolmogorov turbulence, three-dimensional isotropic turbulence (JHTDB), a multiphysics wildfire LES with realistic corrupted measurement operators, and surface-only sensing on a family of unstructured aircraft-wing meshes — evaluated with leakage-audited splits, frozen sensor manifests, per-channel accuracy, proper scoring rules (CRPS, coverage, spread-error), physics metrics (windowed spectra, structure functions), and matched cost accounting (training and inference runtime and peak memory) for every method. Across a fidelity-audited fleet of twelve-plus methods — interpolation and gappy-POD anchors, deterministic operator learners, latent and ambient generative models including DMF-Gen — we find that method ranking is regime-dependent: **[TODO: headline numbers once 2D results land]**. No method recovers unobserved channels of 3D turbulence; every scored ensemble is under-dispersed, and post-hoc recalibration partially repairs it. Benchmark scores are systematically worse than literature-reported numbers on the same data, and we show by controlled ablation that the gap is evaluation protocol — same-trajectory splits, uniform rather than scattered sensing, and favorable reporting — rather than model capability. We release the datasets, splits, sensor manifests, configurations, and evaluation driver as a versioned artifact. **[TODO: finalize abstract numbers and artifact DOI]**

I. INTRODUCTION

Flow fields govern systems that experiments can only observe pointwise: particle-image velocimetry resolves planes of a turbulent flow while the surrounding volume goes unmeasured, flight-test campaigns record surface pressure at taps while the off-body flow that determines performance is inaccessible, and environmental monitoring networks sample the atmosphere at stations kilometers apart. Recovering the full field from such sparse, indirect data is a long-standing inverse problem,^{1–3} and at realistic sensor densities it is severely ill-posed: many fields fit the data, and the missing information sits disproportionately at the small scales. A machine-learning literature now more than a decade deep addresses it with deterministic reconstruction networks,^{4–7} and, more recently, with conditional generative models that sample the distribution of fields consistent with the observations rather than regressing to its mean.^{8–13}

What that literature does not have is a shared evaluation. Community benchmarks for scientific machine learning — PDEBench, The Well, CFDBench, AirFRANS, and their successors — pose forward-surrogate or uniform-downsampling super-resolution tasks, not reconstruction from scattered point sensors; the Johns Hopkins Turbulence Database is a query service with no canonical protocol, and published sparse-reconstruction papers each

cut their own incompatible extraction from it. Three consequences follow, and this paper documents all three. First, *splits leak*: turbulence snapshots are strongly autocorrelated in time, and the standard random-in-time split makes every test frame a near-duplicate of a training frame. On our 3D turbulence data, frame-to-frame correlation is 1.00 at unit separation; under a shuffled split the strongest latent generative baseline scores 0.14 relative L_2 , and under an honest spatially disjoint split the same model scores 0.56 — a factor of four, larger than every method-to-method gap we measure (Sec. VF). Second, *ensembles are drawn but not scored*: generative reconstruction papers routinely show mean-and-standard-deviation maps, several claim calibrated uncertainty on that evidence alone, and, to our knowledge, none reports a proper scoring rule or a calibration diagnostic for 3D single-snapshot reconstruction from scattered sensors;^{14–18} every scored precedent in adjacent settings is under-dispersed. Third, *classical baselines are absent*: training-free interpolation and gappy proper-orthogonal-decomposition floors are rarely reported, and we find one of them (inverse-distance weighting) beats every deep model’s observed-channel error on 3D turbulence at 1% sensor density. A consequence worth stating plainly up front: every number and image in this paper will look worse than its counterparts in the literature, and Sec. VF shows by controlled ablation — same checkpoints, protocols swapped — that this gap is the protocol, not the models.

FlowRecon-Bench is our response: a benchmark for sparse-sensor flow reconstruction built to answer the question a practitioner actually has — *which method family should I use, in which flow regime, at what sensor budget, with what uncertainty and at what cost?* It spans five regimes chosen so that each isolates one axis of difficulty (Sec. II B): laminar cylinder vortex shedding, where the dynamics are low-dimensional and classical linear methods should be near-optimal; two-dimensional Kolmogorov turbulence, the standard testbed of the 2D generative literature;^{8,9} three-dimensional isotropic turbulence under a spatially disjoint evaluation protocol; a multiphysics wildfire LES observed through realistic corrupted operators (noise, occlusion, missing channels); and surface-only sensing on a family of unstructured transonic-wing meshes, where most methods cannot run at all. Every method is evaluated under identical seeded sensor draws, identical augmentation, leakage-controlled splits with a strict tune/test separation, per-channel accuracy, fair CRPS, coverage and spread-error diagnostics, windowed spectral metrics, and — because model selection in practice is a cost-benefit decision — measured training and inference runtime and peak memory for every row.

Our contributions are the benchmark and the findings it forces. (i) A five-regime task suite with leakage-audited splits, frozen sensor manifests, and a fingerprinted evaluation driver, released as a versioned artifact together with per-method configurations and audit notes. (ii) A fidelity-audited fleet of twelve-plus methods spanning classical anchors, deterministic operator learners, latent generative and ambient generative routes, parameter-matched where the published configurations allow and budget-disclosed where they do not (Sec. III). (iii) Findings that generalize beyond any single table: method ranking is regime-dependent; no method recovers channels the sensors do not observe in 3D turbulence (a fleet-wide identifiability limit, not a per-method defect); accuracy and sample realism trade off through the sampler’s operating point, so ranking generative models on mean-field error alone is misleading (Sec. V A); every scored ensemble is under-dispersed, and a post-hoc recalibration fitted on a disjoint tuning split repairs stated coverage (Sec. V C); and the computational axis separates grid methods, whose memory walls we measure, from point methods, whose costs are flat in resolution (Sec. V E).

II. BENCHMARK DESIGN

A. Task

Let $u : \Omega \rightarrow \mathbb{R}^C$ be a flow field drawn from a distribution over fields (snapshots of a turbulent flow, solutions over varying geometry). It is observed through a measurement op-

TABLE I. The five regimes of FlowRecon-Bench. Each dataset isolates one axis of difficulty; the split column names the leakage-controlled protocol (Sec. II C).

Regime	Dataset	Points	Channels	Split	Axis isolated
Laminar periodic	Cylinder wake (new)	2.4×10^4	u, v, p	cross-Re	low intrinsic dim.
2D chaotic	Kolmogorov 256 ²⁸	6.6×10^4	ω	cross-traj.	2D multiscale
3D turbulence	JHTDB isotropic ¹⁹	1.95×10^6	U_x, U_y, U_z, p	spatial	unobs. channels
3D multiphysics	FireBench ²⁰	3.7×10^6	u, v, w, θ, ρ_f	temp. block	corrupted operators
3D geometry	SHIFT-WING ²¹	4×10^5 /case	U_x, U_y, U_z, C_p	by case	surface-only sensing

erator returning readings $\mathcal{O} = \{(\mathbf{x}_j, c_j, y_j)\}_{j=1}^M$ with $y_j = u_{c_j}(\mathbf{x}_j) + \varepsilon_j$, where $\mathbf{x}_j$ is a sensor location, c_j indexes the observed quantity, and ε_j is noise. The task is single-snapshot reconstruction: estimate u — or, for probabilistic methods, sample an approximation to $p(u \mid \mathcal{O})$ — at all query locations, with no access to sensor time histories, initial conditions, or the governing solver. This covers sensors scattered arbitrarily or confined to a surface, observed quantities forming a subset of the target channels, and corrupted readings. Sequential assimilation and forward surrogacy are deliberately out of scope; they are different problems with different information budgets.

B. Datasets: five flow regimes

Each dataset is chosen to isolate one axis of difficulty (Table I); preprocessing details and download/regeneration recipes ship with the artifact.

a. Cylinder wake (new). Two-dimensional incompressible flow past a circular cylinder, generated with OpenFOAM 9 (`pimpleFoam`, laminar) on a 12-block O-grid of 23,800 cells with boundary-layer grading (`checkMesh` clean; maximum non-orthogonality 43.9°). Nondimensionalized by the cylinder diameter and free-stream speed ($D = U_\infty = 1$; $\nu = 1/\text{Re}$), at $\text{Re} \in \{60, 80, 100, 150, 200, 250\}$. Each case runs a transient phase to $t = 120$ and a sampling phase to $t = 270$ written every 0.5 time units — 300 snapshots spanning roughly 25 shedding cycles at $\text{Re} = 100$. Strouhal numbers of the generated cases are verified against established measurements^{22,23} as the physics-level validation of the data. **[TODO: insert measured St table when generation completes.]** The dataset ships in two exports: the native cell-center point cloud (with the wall-adjacent cell ring indexed for a surface-tap sensing variant, the 2D analogue of the wing regime) and a 400×200 uniform-grid interpolation over $x \in [-3, 17]$, $y \in [-5, 5]$ with a body mask, for grid-locked methods — the need for the second export is itself a capability observation. Cross-Re protocol: train on $\text{Re} \in \{60, 100, 150, 200\}$, evaluate on the held-out $\text{Re} \in \{80, 250\}$ (one interpolative, one mildly extrapolative), with temporally blocked splits inside the training cases. An observe- u -only variant probes cross-channel inference (v, p unobserved) in a regime where, unlike 3D turbulence, the phase of the shedding cycle makes the unobserved channels largely determined by the observed one.

b. Kolmogorov turbulence. The two-dimensional forced-turbulence vorticity dataset of Ref. 8, standard in the 2D generative-reconstruction literature: 40 independent trajectories of 320 frames at 256². We stride time by 4 (80 frames per trajectory, 3,200 total) and hold out entire trajectories: 32 for training, 8 for evaluation, assigned by a fixed seeded permutation. Cross-trajectory holdout is the 2D analogue of the spatial holdout below — reconstruction is scored on realizations of the flow the model has never seen at any time.

c. 3D isotropic turbulence. Four 125³ cutouts at disjoint spatial locations of the JHTDB isotropic1024coarse DNS,¹⁹ 50 snapshots each at a stride of 100 database steps, fields (U_x, U_y, U_z, p) , per-field standardized. Training uses cutouts 1–3; all reported evaluation uses cutout 4, spatially disjoint from everything seen in training. Sensors observe U_x and U_z only; U_y and p are never observed, making this the identifiability probe of the suite.

d. Wildfire LES. From the public FireBench store,²⁰ the two large-eddy simulations at 10 and 12 m/s driving wind; we extract a $152 \times 126 \times 192$ region around the fire front ($3.7 \times$

10^6 points; velocity, potential temperature θ , fuel density ρ_f ; 120 snapshots, temporally blocked split with a 10-frame guard gap). Sensors observe the three wind components; the thermochemical channels are inferred. This is the operator-realism regime: evaluation applies noise, slab occlusion, and channel dropout to the sensor sets (Sec. IID).

e. Unstructured wing meshes. SHIFT-WING:²¹ 673 converged RANS cases over transport-aircraft wing-fuselage variants spanning Mach 0.5–0.85, incidence 0.02–3.99°, and 40 geometry parameters. Each case carries 4×10^5 near-body volume nodes and a 6.5×10^4 -node surface pool with pressure and wall shear; observations are surface-only (512 pressure taps + 128 shear gauges drawn from the skin), and the target is the volumetric field. Split 600/73 by case. This regime is the capability filter: methods whose interface cannot accept an unstructured, case-varying point cloud with surface-confined sensors cannot enter without resampling artifacts, and their rows are reported as not applicable rather than silently omitted.

C. Splits and leakage control

Every split in FlowRecon-Bench is chosen against the autocorrelation structure of its data, and the protocol is enforced by tooling rather than convention. Turbulence snapshots correlate at $r = 1.00$ frame-to-frame and $r = 0.67$ at separation 100; random-in-time splits therefore measure near-duplicate interpolation (Sec. VF quantifies the distortion). The suite uses spatial holdout (a disjoint cutout) for 3D turbulence, trajectory holdout for Kolmogorov, Reynolds-number holdout for the cylinder, case holdout for the wing, and temporally blocked splits with decorrelation guard gaps wherever time enters. Model selection is separated from reporting: held-out snapshots are partitioned into a tuning half (odd indices) used for any checkpoint or hyperparameter selection and a test half (even indices) on which every reported number is frozen. The evaluation driver fingerprints the protocol — snapshot indices, sensor counts, and the index-sum of every seeded sensor draw — and aborts on mismatch, so no two methods can silently see different observations. Two incidents during construction illustrate why the tooling exists: an audit found one baseline’s visual-evaluation path silently collapsing the third spatial axis, and the split machinery itself carried a one-frame boundary leak (an `int()` where floating-point arithmetic demanded rounding: at train fraction 0.8, $1 - 0.8$ evaluates below 0.2 and the first held-out-trajectory frame joined the training block). Both were caught by cross-checks the artifact now runs automatically.

D. Sensor protocols and measurement operators

The primary protocol draws sensors uniformly at random over the domain, per observed channel, at densities from 0.1% to 2% of the discretization (regime-dependent; 1% is the headline density), with counts and layouts redrawn per training step for amortized methods and frozen seeded draws at evaluation. Layout variants confine sensors to surfaces (cylinder wall ring; wing skin). Operator variants corrupt the assembled sensor sets by shared seeded code, so every method sees bit-identical corrupted observations: Gaussian noise at $\sigma \in \{0.1, 0.3\}$ in standardized units, contiguous slab occlusion removing 25% of sensors, and channel dropout removing entire observed quantities. Distinguishing per-frame, per-trajectory, and globally fixed sensor masks matters for comparability across papers;²⁴ FlowRecon-Bench uses per-frame draws with frozen evaluation manifests.

E. Metrics

a. Accuracy, per channel. Relative L_2 per channel is primary, reported separately for observed and unobserved channels — aggregates average two different questions and are

shown flagged, never headlined. For ensemble methods both the ensemble-mean error and the single-sample error are reported; the gap between them is the smoothing an ensemble average buys, and single-sample error measures whether individual realizations are physical.

b. Probabilistic scores. For an ensemble $\{s_k\}_{k=1}^K$ and truth y at a point, the fair CRPS estimator¹⁴ is $\text{CRPS} = \frac{1}{K} \sum_k |s_k - y| - \frac{1}{2K(K-1)} \sum_{k \neq l} |s_k - s_l|$, averaged over points and fields in standardized units. The spread-error ratio is the ratio of ensemble standard deviation (RMS over points) to the RMS error of the ensemble mean (ideal 1.0); coverage₉₀ is the fraction of points whose truth falls in the central 90% ensemble interval (ideal 0.90); rank histograms accumulate the rank of the truth within the ordered ensemble. Deterministic methods enter the CRPS column exactly (CRPS reduces to MAE for a point forecast); their dispersion entries are undefined, not zero. Calibration metrics move with the operating point — sampler steps, ensemble size, sensor density — by more than they separate methods, so every calibration number carries its operating point and cross-method comparisons are made only at matched settings. Ensemble size deserves particular care: in a K -sweep on Kolmogorov ($K=8$ –64, same frames and sensors), DMF-Gen’s ensemble-mean rel- L_2 moved only 0.487→0.475 and CRPS was flat (the fair estimator is unbiased), but coverage₉₀ rose 0.59→0.73 — finite-ensemble quantiles are systematically too narrow, so $K=8$ *understates* every generative method’s calibration. We therefore report accuracy and CRPS at $K=8$ and treat $K=8$ coverage as a conservative bound, with the sweep in the artifact.

c. Physics metrics. Shell-averaged energy spectra of single samples (never ensemble means, which destroy small scales by construction), band ratios against the reference solution, structure functions, and velocity-gradient PDFs. One estimator detail is load-bearing enough to state in the protocol: non-periodic cutouts must be windowed before the FFT. On the 3D data the raw spectrum of the *truth* falls only $2.4\times$ from $k=32$ to the Nyquist wavenumber while a variance-preserving Hann taper restores a 5.8×10^5 decay; without the taper, a smooth Gaussian-process draw with zero energy at those wavenumbers registers the same broadband floor as the DNS, and spectral ratios computed there are ratios of two leakage floors that tend to unity for every method. All spectral numbers in FlowRecon-Bench use the windowed estimator.

d. Cost. Model selection is a cost-benefit decision, so cost is a first-class metric family, measured under the same harness for every method on every dataset: training seconds per optimizer step, total GPU-hours to the finished model, peak training memory, inference wall-clock per full-field reconstruction, and peak inference memory. Rows without cost entries are treated as incomplete by the table-assembly tooling. A separate resolution-scaling study (Sec. V E) measures memory versus output discretization from 64^3 to 1024^3 , with out-of-memory walls reported as observed failures, not extrapolations.

F. Fairness methodology

Benchmark tables are only as credible as their weakest baseline, so the fleet is engineered adversarially against our own biases. (i) *Upstream fidelity audits*: every baseline implementation is diffed against its published code before its numbers are admitted, and every declared adaptation (dimensionality ports, set encoders, sampler retuning) is documented in the per-method audit notes; audit incidents (a conditioning defect in the latent flow-matching baseline, sampler-parameter sensitivity in S3GM) were repaired or labelled before reporting. (ii) *Parameter matching*: learned methods are matched to $\sim 6.5\text{M}$ parameters where the published configuration allows; declared waivers (CoNFILd’s published prior at 118.9M) are printed in the table. (iii) *Identical conditions*: the same data files, splits, augmentation, normalization statistics, seeded sensor draws, and metric estimator for every row. (iv) *Budget disclosure*: optimizer steps and wall-clock are reported as run, not silently equalized, and the table notes which budgets exceed which. (v) *Tuning-budget disclosure*: the method we co-developed (DMF-Gen) received more configuration search than any baseline; upstream-faithful configurations are frozen as the headline rows and every



FIG. 1. Per-method capability profile across the deployment axes measured in this paper (filled = yes, half-filled = partial or adapted, open = no). “Operator-robust” and “dissipation-preserving” record the FireBench operator matrix (Table VI) and the windowed dissipation-band ratios (Sec. VD); “memory flat vs resolution” and “sub-second inference” record the measured scaling study (Sec. VE). Axes on which a method was not measured are marked “no” only when the failure is structural; pending measurements are noted in the text.

tuned variant is a separately labelled arm. (vi) *Selection hygiene*: checkpoint selection uses the tuning split only (Sec. IIC).

III. METHODS EVALUATED

The fleet spans four families; Table II records which methods can enter which regimes, itself a benchmark result, and Fig. 1 expands it to the per-method deployment axes — representation freedom, uncertainty, robustness, and scaling — that Sec. V measures. No method fills every column: DMF-Gen alone clears the representation and robustness axes but is the slowest sampler; latent FM inverts that profile exactly. Per-method configurations, adaptations, and audit notes ship with the artifact.

a. Classical anchors. Four training-free floors run on every regime: nearest-neighbor interpolation (KD-tree), inverse-distance weighting over the $k = 8$ nearest sensors, gappy proper orthogonal decomposition²⁵ (POD basis from the training set, modal coefficients by least squares on the sensor readings, rank 80), and the training-mean constant. Distance handling is non-periodic by default — an earlier periodic implementation understated these floors on non-periodic cutouts and was corrected in audit — with periodic variants where the domain genuinely is periodic. As deterministic point forecasts their CRPS equals their MAE. These anchors define what any learned method must beat to justify its training cost.

TABLE II. Capability matrix: which method families can honestly enter which regimes. RS = requires resampling to a grid (lossy near boundaries/meshes); — = interface cannot express the task. [TODO: finalize per-method rows once the 2D fleet lands; consider expanding to one row per method.]

Family	Cyl.	Kolm.	JHU	Fire	Wing
Interpolation/POD anchors	✓	✓	✓	✓	✓
Point-attention det. (Senseiver)	✓	✓	✓	✓	✓
Grid det. (FNO-class)	RS	✓	✓	✓	—
DeepONet-class	✓	✓	✓	✓	✓
Latent-grid generative (LFM)	RS	✓	✓	✓	—
Latent neural-field gen. (CoNFILD)	✓	✓	✓	✓	✓
Voxel/video diffusion (Gen4Turb, S3GM)	RS	✓	✓	✓	—
Point-token generative (SiT-point)	✓	✓	✓	✓	✓
Ambient point-cloud FM (DMF-Gen)	✓	✓	✓	✓	✓

b. Deterministic operator learners. Senseiver⁵ is an attention-based point-native reconstructor mapping arbitrary sensor sets to query values through a latent bottleneck; it trains under the identical sensor randomization as the amortized generative methods. FNO3D places a spectral (FFT-grid) backbone inside the same conditional-generation framework as DMF-Gen, isolating the backbone axis from the formulation axis. DeepONet²⁶ enters in upstream-faithful form (branch/trunk with a declared set-encoder adaptation for variable sensor sets); its near-constant predictions are the pre-registered structural consequence of global mean pooling in the branch, and a structured-branch redesign (“DeepONet++”, multiresolution binned pooling) is reported as a labelled improvement arm. Deterministic models return conditional-mean estimates: optimal in squared error, systematically smoothed, and structurally unable to report what the sensors leave undetermined.

c. Latent generative. Latent flow matching (LFM) is the latent-diffusion-family standard:^{11,27} a convolutional autoencoder trained on full fields, followed by conditional rectified flow in its latent space with a point-encoder conditioning branch. It is the strongest aggregate performer on 3D turbulence and carries the family’s structural signature, a dissipation-range spectral floor inherited from the autoencoder.^{28,29} CoNFILD¹¹ is the published latent neural-field competitor: a conditional-neural-field autodecoder compressing each snapshot to a latent vector, an unconditional diffusion prior over latents, and diffusion posterior sampling for conditioning; three arms are reported (published prior; capacity-matched; 384-dimensional), with the latent-capacity ceiling of the autodecoder measured explicitly as the decoder bound.

d. Ambient generative. Gen4Turb¹² is voxel diffusion: an elucidated-diffusion 3D U-Net with mask-channel conditioning, run at its native 120³ on the 3D data and retrained under the benchmark’s observation model (the most generous variant is reported). S3GM⁹ is a pretrained video-diffusion prior with sampling-time guidance, natively two-dimensional; its 3D entry is a declared adaptation treating one spatial axis as video time, and its published fixed guidance weights required tuning-split renormalization to converge at 3D scale. SiT-point adapts a scalable interpolant transformer³⁰ with a point tokenizer (not patchification): sensors and queries are tokenized directly, generation runs in fixed-size token chunks, and the chunk budget is the visible constraint (member-level chunk-incoherent speckle that ensemble averaging hides — one reason FlowRecon-Bench reports member-level metrics). DMF-Gen³¹ performs conditional rectified flow directly on field values at scattered points — ambient function space, no autoencoder, no voxelization — with a Gaussian-process source shared across discretizations, conditioning factored into a global cross-attention summary plus a local k -nearest-sensor retrieval per query, a Monte-Carlo subsampled training objective (a random $\sim 2\%$ of points per step), and training amortized over sensor counts, layouts, noise, and observed-channel subsets. We evaluate it as published, at the benchmark’s pa-

parameter budget; it is one row of the fleet, not the paper’s contribution.

IV. RESULTS BY REGIME

A. Laminar vortex shedding: where classical methods should suffice

The dataset was generated for this benchmark (Sec. II B) and verified against the literature before any model saw it: Strouhal numbers 0.140/0.160/0.173/0.187/0.200/0.207 for Re 60–250 track the Williamson–Norberg correlations, mean drag sits where the 6% blockage predicts, and lift amplitude grows monotonically with Re . Sensors observe (u, v) at 1% of the 23,800 mesh cells (238 per field); pressure is never observed; evaluation is on the held-out Reynolds numbers $\{80, 250\}$.

The classical floors decide this regime before the learned fleet reports. The training family’s POD spectrum is essentially exhausted at rank 20 (99.7% energy), and *gappy POD with 20 modes reconstructs held-out Reynolds numbers at 0.056 relative L_2 — including the never-observed pressure at 0.05* — because the modal coefficients inferred from sparse velocities carry the full coupled state. Pure interpolation cannot do this (p column: 1.0 by construction) and trails badly on the velocities as well (IDW 0.71 aggregate). This is the mirror image of 3D turbulence, where the same POD basis was useless (0.95, Table IV) and interpolation was the strong floor: *which classical method is the one to beat is itself regime-dependent*, and a learned method only earns its complexity here if it approaches the POD floor at comparable cost. [TODO: learned-fleet rows (training released after the smoke gate), cross- Re breakdown, surface-tap variant, ensemble-collapse (spread) check for the generative rows.]

B. Two-dimensional Kolmogorov turbulence

Sensors observe vorticity at 1% of the 256^2 points (655 sensors); learned models are scored on 50 evenly spaced frames of the eight held-out trajectories ($K=8$, identical canonical seeded draws across methods; classical floors on all 640). Table III reports the first learned rows alongside the floors, and Fig. 4 (left) the sensor-count sweep. Two regime findings stand out. First, *the 3D ranking does not transfer*: the ambient point methods DMF-Gen (0.487 rel- L_2 , 0.235 CRPS) and SiT (0.494, 0.234) are statistically tied at the front and both beat inverse-distance weighting (0.546), while latent FM — the 3D aggregate winner — trails the interpolation floor here (0.600) at the matched $\sim 50k$ -step budget. Second, the floors carry regime information of their own: gappy POD’s rank-80 training basis captures only 70.5% of the training-set energy — two-dimensional turbulence is not low-rank in the POD sense, unlike the cylinder — and plain inverse-distance weighting beats it by 20% relative L_2 . [TODO: Senseiver/MLP-RBF/GeoFNO/S3GM rows (training/eval in queue); paired CIs; spectra panel; operator variants ported down from the FireBench matrix.]

C. Three-dimensional isotropic turbulence: the full fleet

The 3D turbulence benchmark is the controlled comparison: a periodic-topology regular grid is native terrain for voxel, patchified, and latent-grid methods, so every family competes at full strength. Sensors observe U_x and U_z at 1% of 1.95×10^6 points per channel; U_y and p are never observed. Table IV reports per-channel accuracy and probabilistic scores for the full fleet on all 50 held-out snapshots of the spatially disjoint cutout under canonical seeded draws; Table V reports the matched cost accounting.

The honest headline is that no method leads every column. SiT-point posts the best observed-channel errors (0.143/0.132) at roughly 150s per field — 239 sequential 8,192-token chunks at 32 steps — and its single samples carry chunk-incoherent speckle that

TABLE III. Kolmogorov 256², trajectory-holdout evaluation (vorticity observed at 1% = 655 sensors, identical canonical seeded draws; learned rows: 50 evenly spaced held-out frames, $K=8$; classical floors: all 640). Inference cost per field on one H100. [TODO: remaining fleet rows; ens. vs single rel- L_2 split; paired CIs.]

Method	Rel. L_2	CRPS	Cov.90	s/field	GB
DMF-Gen (NFE 4)	0.487	0.235	0.59	0.19	0.76
SiT (50 steps)	0.494	0.234	0.64	0.18	0.24
Latent FM (NFE 4)	0.600	0.310	0.64	0.03	0.19
IDW ($k=8$, periodic)	0.531	0.375	—	—	—
IDW ($k=8$, non-per.)	0.546	0.385	—	—	—
Nearest neighbor (per.)	0.582	0.371	—	—	—
Gappy POD ($r=80$)	0.680	0.511	—	—	—
Training mean	1.000	0.753	—	—	—

TABLE IV. 3D isotropic turbulence, spatially disjoint evaluation region (1.95M points; U_x, U_z observed at 1% per channel, U_y^*, p^* unobserved; canonical seeded sensor draws, all $n=50$ held-out snapshots, $K=8$ ensembles at each method’s listed sampler setting). Per-channel relative L_2 of the ensemble mean is primary; $\text{agg}^\dagger$ averages all four channels and is flagged, not headlined. Unobserved-channel error ≥ 1.0 is worse than predicting the training mean; values near 1.0 across the fleet reflect a shared identifiability limit of point-conditioned methods (Sec. VB). Spread-error and coverage ideals are 1.0 and 0.90; for deterministic methods CRPS reduces to MAE and dispersion is undefined. Step budgets are as-run, not equalized (Table V). Labelled improvement arms (Senseiver capacity, DeepONet++, CoNFILd repairs, S3GM normalized guidance) are reported in the artifact. [TODO: S3GM canonical row finalizing; CoNFILd strict-capacity arm completed 2026-09-03 (rel- L_2 0.90) — fold in with its arm label.]

Method	U_x	U_y^*	U_z	p^*	$\text{agg}^\dagger$	CRPS	Spr/Err	Cov.90
DMF-Gen (NFE 4)	0.176	1.048	0.182	0.964	0.593	0.291	0.63	0.54
Latent flow matching	0.248	0.727	0.257	0.646	0.469	0.244	0.68	0.56
SiT-point ($N=32$)	0.143	0.968	0.132	0.937	0.545	0.338	0.75	0.58
FNO3D (same framework, NFE 4)	0.188	0.904	0.191	1.061	0.586	0.269	0.94	0.63
CoNFILd (published prior, DPS 1000)	0.409	0.751	0.374	1.007	0.635	0.371	0.82	0.57
CoNFILd (capacity-matched)	0.488	1.089	0.420	1.487	0.871	0.562	0.62	0.45
CoNFILd (384-d)	0.524	0.604	0.451	1.401	0.745	0.480	0.69	0.50
Gen4Turb (anneal, 32-step)	0.224	0.996	0.455	1.005	0.670	0.407	0.37	0.33
S3GM (JHU-tuned, $N=200$)	sampler diverges at 3D scale (rel- $L_2 \sim 10^6$); see text							
Senseiver (deterministic)	0.361	1.077	0.380	0.978	0.699	0.479	—	—
DeepONet (deterministic)	0.593	0.972	0.652	0.985	0.800	0.570	—	—
Nearest neighbor (KD-tree)	0.209	1.000	0.219	1.000	0.607	0.406	—	—
IDW ($k=8$, non-periodic)	0.174	1.000	0.181	1.000	0.589	0.396	—	—
Gappy POD ($r=80$)	0.523	1.306	0.540	1.443	0.953	0.673	—	—
Training mean	1.000	1.000	1.000	1.000	1.000	0.786	—	—

the ensemble mean averages away. Latent flow matching leads the aggregate, the unobserved channels, and CRPS: its convolutional decoder emits globally structured fields even where no sensor constrains them, at the dissipation-range price quantified in Sec. VD. Non-periodic inverse-distance weighting — a training-free interpolant — edges the best learned observed-channel error (0.174 vs. 0.176 on U_x for DMF-Gen), so no learned method beats interpolation on observed channels at 1% density; the learned and generative payoffs concentrate where the problem is ill-posed — unobserved channels, corrupted operators (Sec. IV D), sparser sensing — and in sampling cost (DMF-Gen reaches its accuracy in 4 network evaluations against 32–1000 for the guided-diffusion and transformer baselines). DeepONet’s near-constant predictions are the structural consequence of global mean pool-

TABLE V. Cost accounting for the 3D-turbulence fleet (H100-SXM 80 GB). Steps are as-run, not equalized: every external baseline received at least DMF-Gen’s 48k optimizer steps except SiT-point (wall-clock-matched, 38.5k); CoNFILD stages train on a wall-clock budget. “(d)” = derived from evaluation-job wall time rather than a dedicated benchmark. [TODO: unify derived vs. dedicated-bench inference timings.]

Method	Params	Opt. steps	Train s/step	Train GPU-h	Train peak GB	Infer s/field
DMF-Gen (NFE 4)	6.51M	48k	0.574	7.7	53.6	~17 (d)
Latent flow matching	6.68M	209k+95k	0.238	13.9	3.6	0.019
SiT-point ($N=32$)	6.57M	38.5k	1.81	19.4	26.8	~150 (d)
FNO3D (NFE 4)	6.73M	79.9k	0.766	17.0	48.6	0.097
CoNFILD (pub. prior)	118.9M ^w	wall	35.5/ep	~19	4.7	~130 (d)
Gen4Turb (anneal)	6.16M	105k	0.598	17.4	47.4	1.65
S3GM (JHU-tuned)	6.35M	105k	0.635	18.5	26.3	~46
Senseiver	6.42M	79.2k	0.776	17.0	22.4	<1
DeepONet	6.51M	82.0k	0.29	6.7	15.7	0.23
IDW $k=8$ (CPU)	—	—	—	—	10 (RSS)	0.68
Gappy POD $r=80$ (CPU)	—	45 s fit	—	—	10 (RSS)	0.14

TABLE VI. Wildfire LES (3.7M points, 5 fields) under realistic measurement operators (relative L_2 of ensemble mean / fair CRPS; $K=8$, 8 snapshots, identical seeded operator realizations for all methods). Sensors observe the three wind components at ~1% of points; θ and ρ_f are never observed. DMF-Gen is a single amortized model across all rows. The noise- 0.3σ row is out of distribution for it (training noise reached 0.1σ). Baseline cells are the frame-matched re-evaluation (identical absolute frames and seeded operator realizations for all methods; a frame-index audit found the original baseline evaluation used an earlier window of the same held-out tail and mildly flattered the baselines — see the released audit trail).

Operator	DMF-Gen		Latent FM		Senseiver	
	Rel. L_2	CRPS	Rel. L_2	CRPS	Rel. L_2	CRPS
Clean	0.456	0.234	0.719	0.482	0.840	0.608
Noise $\sigma=0.1$	0.458	0.237	0.718	0.482	0.840	0.609
Noise $\sigma=0.3$	0.461	0.248	0.721	0.484	0.840	0.610
Slab occlusion (25%)	0.511	0.275	0.731	0.491	0.840	0.609
Channel dropout (u only)	0.603	0.355	0.776	0.474	0.923	0.671

ing in its branch; the labelled DeepONet++ arm (structured pooling) improves U_x from 0.593 to 0.512 and remains behind the point-attention family. [TODO: paired per-snapshot CIs/TOST for the top-3 comparisons from the canonical $n=50$ JSONs; seed replicates (running) for the DMF-Gen row.]

D. Multiphysics LES under corrupted operators

Table VI isolates operator realism on a regular LES extraction (grid methods participate on their native terms). The amortized point-cloud model wins every cell, and the reading of the baselines’ apparent stability matters more than the win: their numbers barely move across rows because they are bias-dominated — a model that extracts little of the information in the sensors loses little when that information is corrupted. The latent baseline reconstructs at 0.72–0.78 relative L_2 across all operators (predicting zero scores 1.0), and its CRPS *improves* slightly under channel dropout only because its spread is large where its mean is wrong. Graceful monotone degradation, not invariance, is the behavior a practitioner should want from a reconstruction system, and only the model that uses the sensor

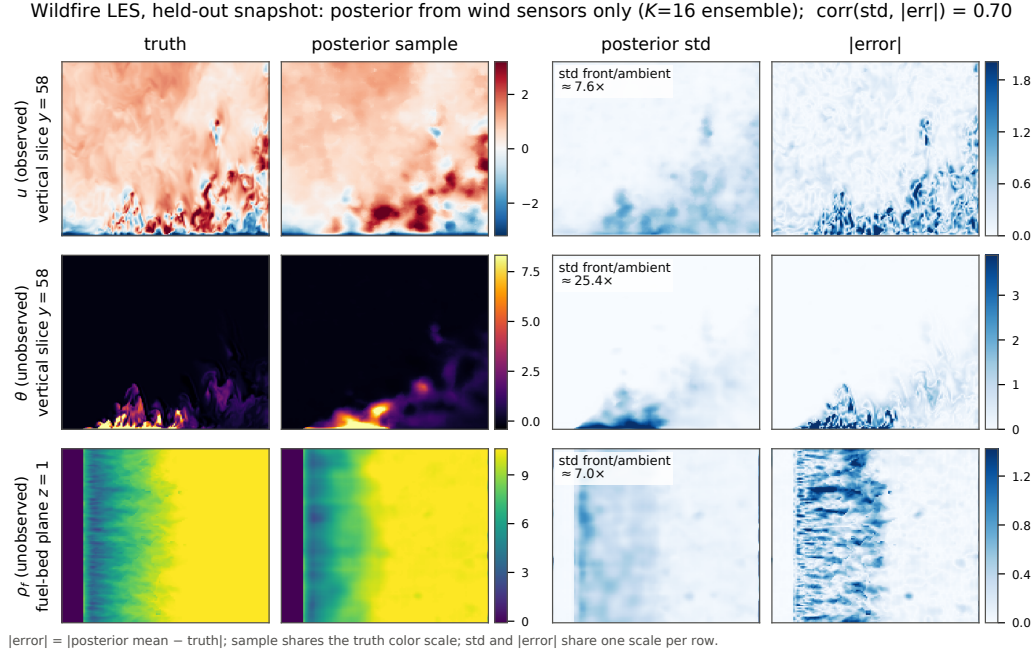


FIG. 2. Wildfire LES, held-out snapshot, conditioned on wind sensors only (DMF-Gen). Wind and potential temperature on a vertical slice through the plume; fuel density on the fuel-bed plane. Potential temperature and fuel density are never observed; from wind alone the model localizes the burn scar, and the predictive standard deviation concentrates exactly on the front.

information exhibits it. A clean-trained control separates the mechanisms: it matches the operator-amortized model under noise and occlusion (robustness there is architectural) but collapses under channel dropout (0.743 vs. 0.603), the one operator that changes the structure of the observation set — and operator-amortized training costs nothing in the clean condition (0.455 vs. 0.456). Cross-variable inference falls out of the same interface: from wind alone, unobserved fuel density is recovered at 0.044 relative L_2 (fire-front localization) and potential temperature at 0.49, and the predictive standard deviation concentrates on the burn front — 7–25 $\times$ its ambient value over the hottest 5% of the domain — which the model never observes (Fig. 2). [TODO: add one voxel-diffusion row (Gen4Turb-class) for route completeness per the gap analysis.]

E. Unstructured geometry with surface-only sensing

The wing is the capability filter: an unstructured mesh whose geometry varies across cases, and observations confined to the body surface. Voxel diffusion, patchified transformers, and latent-grid flow matching cannot be trained here without resampling the adaptive mesh onto an axis-aligned grid — a lossy step that erases exactly the near-wall resolution the mesh exists to provide — so their rows are reported as *not applicable without resampling* rather than silently omitted; point-native methods (Senseiver, CoNFILd-class neural fields, SiT-point, gappy POD on the mesh, DMF-Gen) compete fairly. From 640 surface sensors (512 pressure taps, 128 shear gauges, Mach and incidence as parameter tokens), DMF-Gen recovers the volumetric pressure-coefficient field on held-out geometries at 0.126 relative L_2 and velocities at 0.53–0.69 (CRPS 0.18, spread-error 0.68, coverage 0.38; $K=4$, NFE 4), with predictive standard deviation concentrating in the boundary layer and wake — the regions genuinely undetermined by surface data (Fig. 3). [TODO: wing baseline table (Senseiver, SiT-point, MLP-RBF, gappy POD, IDW rows + n/a rows) — the runs

Transonic wing, held-out geometry: volume reconstruction from 512 surface pressure taps + 128 shear gauges (midspan slice); $\text{corr}(\text{std}, |\text{err}|) = 0.77$

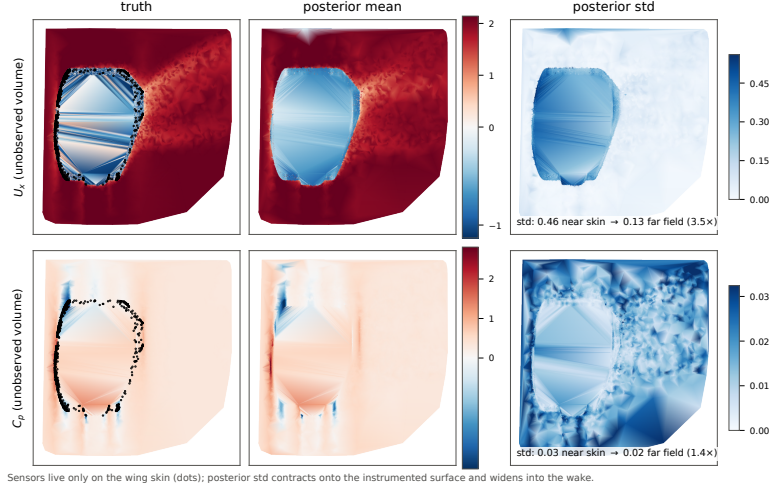


FIG. 3. Wing, held-out geometry, midspan slice: volumetric reconstruction from surface pressure taps and shear gauges alone (DMF-Gen). The predictive standard deviation is largest in the boundary layer and wake, and spread and error decay together away from the skin (pointwise correlation 0.77).

are inventoried in the gap analysis and queue behind the 2D fleet.]

V. CROSS-CUTTING ANALYSES

A. The distortion–perception tradeoff, operationally

Relative L_2 of the ensemble mean is minimized by the posterior mean — by construction it rewards collapse toward a smooth conditional average — while proper scores and physics metrics reward calibrated diversity and sharp members. The benchmark measures the tension directly through the sampler’s operating point: for the rectified-flow methods, two integration steps give the best mean-field error and the sharpest single samples, four steps are the CRPS optimum, and dispersion rises monotonically with further steps at flat CRPS (spread–error 0.65 at NFE 2 to 0.92 at NFE 16 for the $\lambda_{\text{sp}}=0$ DMF-Gen variant). A spectral training penalty moves along the same frontier: it buys single-sample sharpness and inertial-band fidelity at a measured cost in dispersion. The practical rule the benchmark enforces: never rank generative reconstruction methods on mean-field error alone, and attach the sampler operating point to every probabilistic number.

B. Unobserved channels: a fleet-wide identifiability limit

On 3D turbulence, no method in the fleet reconstructs the unobserved U_y below 0.60 relative L_2 , and most sit near or above 1.0 — worse than predicting the training mean (Table IV). The two partial exceptions (latent FM at 0.727, CoNFILD-384d at 0.604) are the two methods whose decoders impose the strongest global field structure, and both pay for it elsewhere (dissipation-range truncation; observed-channel accuracy). This is not a per-method defect: at 1% point sensing on two velocity components, the instantaneous realization of a third component of 3D turbulence is close to unidentifiable, and only its statistics are recoverable. The observation disciplines how such results should be reported — per channel, with unobserved channels flagged — and motivates the cylinder’s observe-

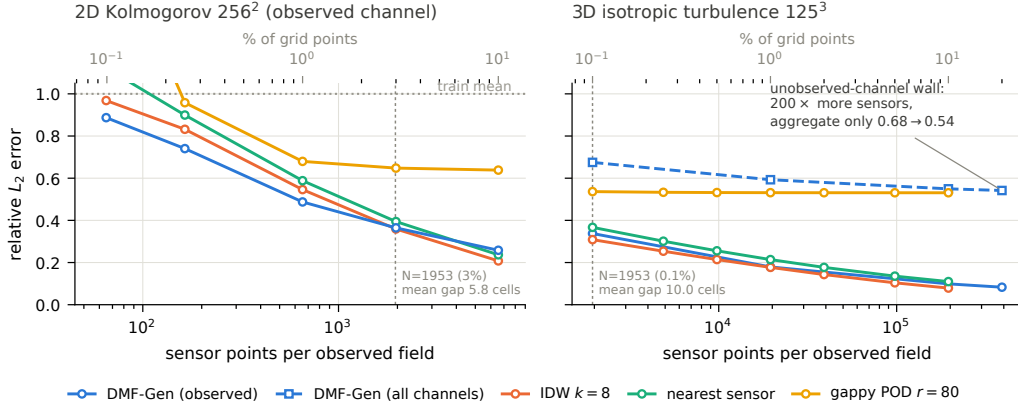


FIG. 4. Reconstruction error versus the *number* of sensor points, in 2D (left, Kolmogorov 256^2) and 3D (right, JHU 125^3 cross-cube; solid = observed-channel $\text{rel-}L_2$, dashed = aggregate over all four channels). The two panels sit on opposite sides of the density story: the same sensor count that leaves a 5.8-cell mean gap in 2D leaves a 10-cell gap in 3D, and while every observed-channel error falls steadily with sensor count — DMF-Gen tracking the interpolation slope from 0.33 to 0.09 over $200\times$ — the 3D *aggregate* saturates near 0.54 because the unobserved channels do not identify (Sec. VB). Gappy POD is flat at every density: 3D isotropic turbulence has no low-rank structure for added sensors to project onto. DMF-Gen curves: $K=8$, NFE 4, 50 held-out snapshots; classical curves: same seeded sensor draws. [TODO: 2D learned-fleet curves when training completes.]

u -only variant, where the shedding phase *does* determine the unobserved channels and the same test separates methods rather than hitting a physics ceiling. [TODO: cylinder cross-channel numbers when the 2D fleet lands.]

C. Calibration: characterized, then repaired

Every scored ensemble in the fleet is under-dispersed at the headline operating point (spread-error 0.37–0.94 against the ideal 1.0; coverage 0.33–0.63 against 0.90), consistent with every scored precedent in adjacent literatures.^{15,16} For the best-characterized model the miscalibration has a low-dimensional mechanism: predictive spread follows a distance-to-nearest-sensor profile with a hard floor (≈ 0.09 in standardized units), the profile is density-invariant, and the sampler step count rescales it wholesale ($\approx 1.5\times$ from NFE 4 to 16). That structure makes post-hoc repair tractable, and the benchmark ships two estimators fitted strictly on the tuning split and frozen on test: a per-density scalar spread multiplier, and conformalized quantile scaling binned by distance-to-nearest-sensor, which carries a finite-sample marginal coverage guarantee on exchangeable points. The identical wrapper is offered to the generative baselines, so the repaired comparison is not one-sided. Three honesty requirements govern the reporting: the repair is post hoc and stated as such; it rescales stated uncertainty and cannot sharpen the spread floor, which remains the residual limitation; and all repaired numbers carry their operating point. [TODO: TEST-split before/after spread-error and coverage for both estimators, DMF-Gen and latent FM, at 2–3 densities (fits exist: `src/recalibrate_spread.py`, `src/conformal_recalib.py`; per-point dumps queued).]

D. Spectral fidelity splits by band

Windowed shell spectra of single samples (Fig. 5) separate the generative families where pointwise error cannot. On 3D turbulence the latent route holds more inertial-range en-

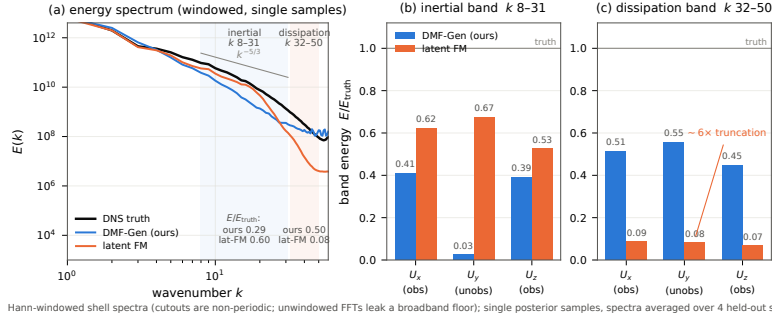


FIG. 5. Turbulence statistics of single posterior samples on the held-out 3D region (4 snapshots, NFE 4, Hann-windowed; ensemble means are deliberately not shown — averaging destroys small scales by construction). Left: shell-averaged energy spectra — the latent baseline tracks the DNS through the inertial range then falls away sharply in the dissipation band (autoencoder floor); the ambient model sits lower in the inertial range and flattens into a noise plateau at high k . Middle: longitudinal structure functions. Right: velocity-gradient PDFs.

ergy than the ambient point-cloud route (0.62 vs. 0.41 of DNS energy on the observed U_x , $k \in [8, 31]$) but collapses in the dissipation band, retaining 0.07–0.09 of DNS energy over $k \in [32, 45]$ against the ambient model’s 0.40–0.46 — the autoencoder spectral floor^{28,29} appearing where it was predicted, a factor-of-five separation invisible to relative L_2 . Neither family reproduces the DNS spectrum from 1% sensors: the ambient model’s high- k content flattens into a broadband-noise plateau on the unobserved pressure channel (3.5–4.8 $\times$ DNS energy), and on channels no sensor constrains its samples stay near its spectrally empty source prior (inertial ratio 0.03–0.06 on U_y , versus 0.67 for the latent decoder that paints texture everywhere). Member-level spectra, windowed estimation, and per-channel reporting are all load-bearing; ensemble-mean spectra and unwinded estimators each silently manufacture different conclusions.

E. Computational cost and scaling

Cost separates the fleet along two axes. At the benchmark resolution (Table V), total training cost to a finished model spans 7–19 GPU-hours across learned methods — comparable — but inference spans five orders of magnitude, from 19 ms (latent decode) to minutes (chunked token generation, DPS optimization), and peak training memory spans 3.6–54 GB. Against resolution (Fig. 6), the fleet splits into two families: grid computation (voxel diffusion, convolutional autoencoders) scales cubically with measured out-of-memory walls at 320^3 (training, batch 1) and 640^3 , while point computation (attention over sensors with chunked queries) is flat to 1024^3 in both training and inference memory. The honest converse is that grid computation is cheaper below the crossover ($\sim 240^3$ for inference: a convolutional decode of a 125^3 field costs 6 ms where chunked point evaluation costs seconds). No family dominates; the benchmark’s contribution is that the walls and crossovers are measured, so a practitioner can place their discretization on the axis before choosing a method family.

Figure 7 makes the same point from the accuracy side: on the cost–accuracy plane alone the latent baseline dominates every method including DMF-Gen, and the case for ambient point-cloud generation rests entirely on the axes the plane omits — resolution scaling, geometry, operator robustness, and dissipation-range fidelity.

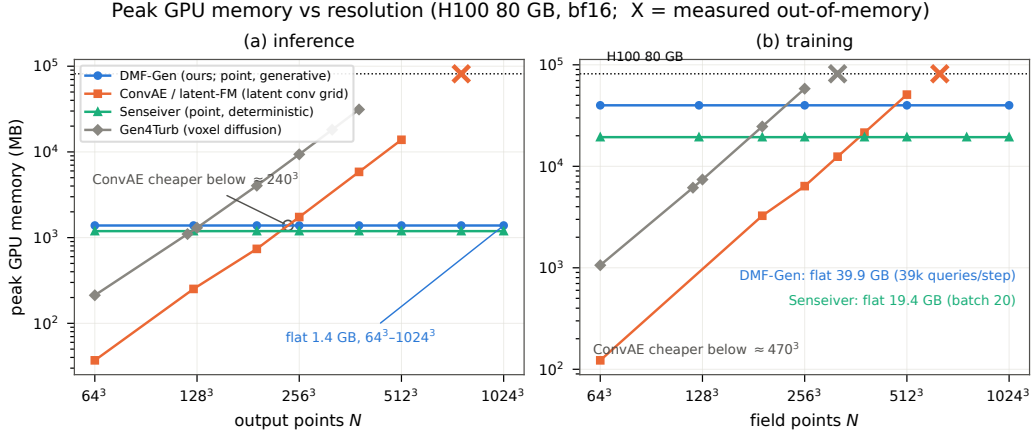


FIG. 6. Measured peak memory versus output resolution (H100, bf16, log-log; OOM markers are measured failures, not extrapolations) for full-field sampling (left) and training steps (right). Grid methods (voxel diffusion, convolutional AE) scale cubically and fail at 320^3 and 640^3 ; point methods (DMF-Gen, Senseiver) are flat to 1024^3 (10^9 points). Annotation: fraction of the field supervised per training step — 100% forced for grid methods, a fixed 39k-point budget for the subsampled point method.

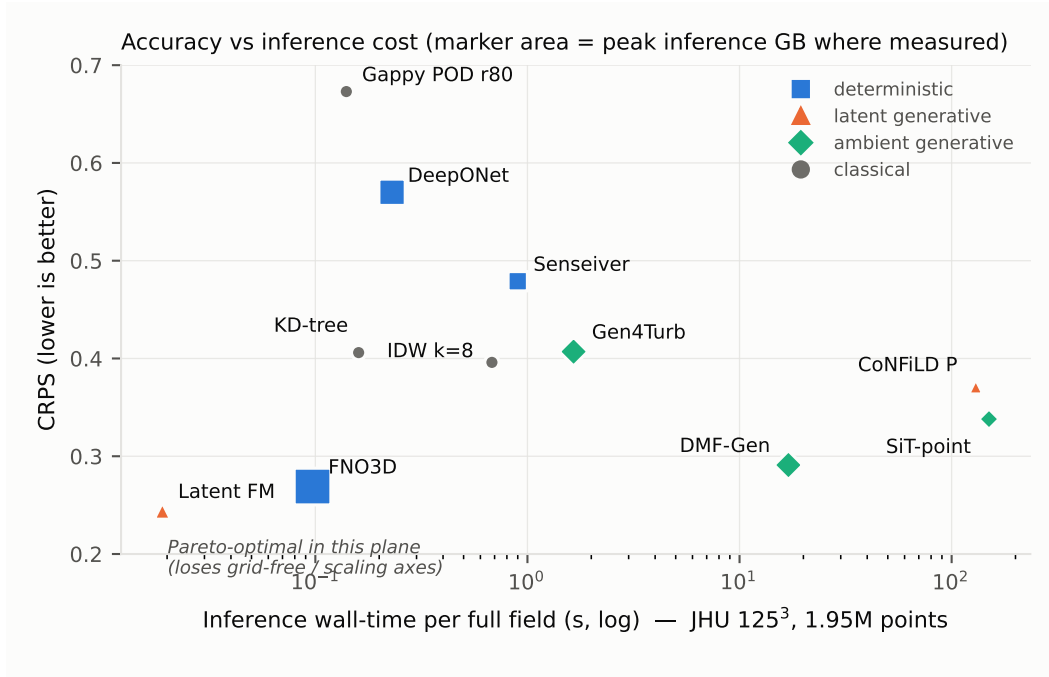


FIG. 7. Probabilistic accuracy (CRPS, JHU cross-cube, $n=50$) against inference wall-time per full 125^3 field, marker area proportional to peak inference memory where measured (Table V). At the benchmark resolution the latent flow-matching baseline Pareto-dominates this plane outright — fastest and lowest CRPS — which is precisely why a single cost-accuracy plot is an insufficient basis for method choice: the axes it cannot show (Fig. 1; the scaling walls of Fig. 6; the operator matrix of Table VI) are where the grid-locked methods lose. S3GM is absent: its upstream sampler diverges at 3D scale (Sec. IV C).

F. Why benchmark numbers look worse than the literature’s — protocol, not architecture

Readers who compare our tables and galleries against published reconstruction papers — including papers evaluated on the very same Kolmogorov data⁸ — will notice that every method here, including the strong ones, scores and *looks* worse. We consider this observation a central result of the benchmark rather than an embarrassment, because each contribution to the gap is a protocol choice, and we measure them separately.

(i) *Splits: interpolation vs. generalization.* On 617 consecutive DNS frames, frame-to-frame correlation is 1.00 at unit separation and 0.67 at separation 100, and a shuffled random-in-time split places every evaluation frame within one frame of a training frame. Under that split the latent flow-matching baseline scores 0.14 relative L_2 ; under the spatially disjoint protocol the same architecture, data, and budget scores 0.56 — and the entire fleet shifts by factors of 2–4, unevenly, so leakage does not merely inflate scores, it *reorders methods*: models with the capacity to memorize the training trajectory (latent codecs; sensor-identified frame lookup) gain most. The 2D fleet reproduces the mechanism: Senseiver reaches 0.17 training loss but 0.91 held-out-trajectory error on Kolmogorov — the same memorization that a same-trajectory split would have rewarded as excellence. The controlled 2D ablation sharpens the mechanism: the *same* DMF-Gen checkpoint evaluated on unseen frames of *seen* trajectories scores 0.483 against 0.487 held-out — no gain, because this model never memorized (its train and validation losses track). Leakage is selective: it pays only models with the capacity and incentive to memorize, which is precisely why bad splits *reorder* rankings instead of lifting all methods equally. [TODO: Senseiver protocol-B cell (expected to collapse from 0.913 toward its 0.17 train loss) + 3-method shuffled-vs-honest table from the archived pre-fix evaluations.]

(ii) *What “1% sparsity” means.* Much of the visually striking literature conditions on uniformly downsampled coarse grids — every cell of a coarse lattice, i.e. band-limited full-field information with bounded gaps — or adds physics guidance (PDE-residual or forcing terms at sampling time^{8,10}). Our protocol uses randomly scattered points and purely data-driven conditioning: the same nominal percentage carries far less information and admits large sensor-free regions. Measured on Kolmogorov with the same checkpoint and frames, rearranging the 655 scattered sensors onto a uniform 25×26 lattice improves DMF-Gen from 0.487 to 0.367 relative L_2 (CRPS 0.235→0.171) and IDW from 0.546 to 0.452 — a 25% accuracy difference from sensor *layout* alone, before any change of model or split.

(iii) *What is shown.* Publication figures typically display ensemble means or favorable frames; we display single posterior samples on fixed, pre-registered frames (our gallery frame runs 0.65–0.96 across the fleet against 0.39–0.50 fifty-frame averages), because single-sample behavior is what the physics metrics score. Ensemble size interacts with the same point: $K=8$ coverage understates calibration (Sec. II E), and ensemble means visually erase exactly the small scales the spectra measure.

(iv) *Matched budgets.* Every model trains at a matched ~ 50 k-step budget with disclosed tuning asymmetry, not to per-method convergence. Fairness costs polish; the cost table prices it.

The synthesis we intend a reader to take away: *apparent reconstruction quality in this literature is substantially a property of evaluation protocol, and only secondarily of architecture.* A benchmark that fixes the protocol first makes the remaining — genuine — architectural differences measurable.

VI. DISCUSSION

a. *Which method should a practitioner use?* The benchmark’s answer is conditional, and that is the point. If sensing is dense relative to the flow’s intrinsic dimension (laminar, periodic regimes) use interpolation or gappy POD; nothing learned earns its training cost there. [TODO: confirm with cylinder numbers.] On observed channels of turbulence at $\sim 1\%$ density, training-free interpolation remains competitive with the best learned meth-

ods — the case for learning begins where the problem is ill-posed: unobserved channels, corrupted or surface-confined operators, and sparser sensing. There, deterministic learners are structurally limited to conditional means; among generative routes, latent compression buys inference speed and global field structure at a measured dissipation-range floor, while ambient point-cloud generation buys geometry-freedom, operator amortization, and small-scale fidelity at higher inference cost. If the discretization exceeds $\sim 300^3$ or lives on an unstructured mesh, grid families exit on measured walls or interface grounds. Uncertainty from any current method must be recalibrated before operational use, and the benchmark ships the wrapper.

b. Limitations. All learned rows train per-regime (no pretraining across flows); the suite contains no experimental data — particle-image-velocimetry or pressure-sensitive-paint regimes with real sensor noise are the natural sixth regime; single-snapshot reconstruction ignores temporal information that assimilation exploits; the cost accounting is single-GPU (H100) and does not probe multi-GPU scaling; the tuning-budget asymmetry of Sec. IIF item (v) is disclosed but not eliminated; and training-seed variance is measured for only part of the fleet. [TODO: seed-replicate numbers when the replicate chains finish.]

VII. CONCLUSION

FlowRecon-Bench turns sparse-sensor flow reconstruction from a collection of incompatible per-paper evaluations into a measured landscape: five regimes, twelve-plus audited methods, proper probabilistic scoring, physics metrics with their estimator pitfalls documented, matched cost accounting, and leakage-controlled splits with the distortion of the dominant protocol quantified. The findings that outlive any single table: method ranking is regime-dependent; interpolation is the floor to beat and sometimes is not beaten; unobserved channels of 3D turbulence are near-unidentifiable for every method; mean-field error alone misranks generative methods; every scored ensemble is under-dispersed and post-hoc recalibration repairs stated coverage; and the computational axis separates grid from point families at measured walls. The datasets, splits, sensor manifests, configurations, audit notes, and evaluation driver are released for extension — the leaderboard is meant to be beaten. [TODO: tighten once 2D results land.]

ACKNOWLEDGMENTS

[TODO: funding/compute acknowledgments; DMF-Gen authors; JHTDB, FireBench, SHIFT-WING dataset providers.]

DATA AVAILABILITY

The benchmark artifact — datasets or regeneration recipes (JHTDB cutout coordinates and download scripts, OpenFOAM case definitions, converter scripts), split manifests, seeded sensor draws, per-method configurations, audit notes, and the evaluation driver — will be released on acceptance. [TODO: Zenodo/HF DOI; confirm JHTDB redistribution terms (fallback: recipe + coordinates); needs sign-off before anything goes public.]

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